Smooth Einstein–scalar perturbations through timelike shell contact

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Abstract

We construct smooth linearized Einstein–scalar perturbations that evolve through contact with a timelike shell in a fixed five-dimensional model. A conformal map sends the hyperboloidal shell to the equator of a round hemisphere and converts its exact even boundary law into a dynamic boundary equation with positive auxiliary inertia; the odd law becomes Robin. For finitely many angular harmonics and arbitrary smooth radial collar data, a characteristic integral system coupled to a boundary ordinary differential equation proves existence, uniqueness and smoothness on an explicit finite slab. Time reversal and one-dimensional sideways uniqueness prove actual boundary activation for pure displacement or pure velocity data. Direct graph geometry then verifies all metric and scalar junctions, including the odd shell displacement. The action-derived covariant current requires a moving-shell cut term in odd parity. A radial Poincaré bound gives a nondegenerate physical Cauchy pairing on a sufficiently thin compact annulus and a nonzero total pairing conserved through contact. No nonzero local shell current, general completed phase space, nonlinear stability or cosmological origin claim is inferred.

Status. Technical continuation for author review. This is not an APS submission or external peer review. Detailed derivations and executable identity checks accompany the draft; verification scope and declarations appear in Section 8.

1. Fixed model and the statement proved

Use signature $(-++++)$, $8\pi G_5 = 1$, and two interior copies joined at a common timelike shell:

$$S = \sum_{i=\pm} \left\{ \int_{M_i} \epsilon_g [R/2 - (\nabla\phi)^2/2 - U(\phi)] + \int_\Sigma \epsilon_h K_i \right\} - \int_\Sigma \epsilon_h \sigma(\phi). \tag{1}$$

Both outward normals point toward increasing interior radius. The fixed registered functions are

$$\begin{aligned} &\eta = \phi - 1, \quad W = \tfrac{1}{3} + \eta^2 + \tfrac{1}{3}\eta^3, \quad U = \tfrac{1}{2}W_{,\phi}^2 - \tfrac{2}{3}W^2, \\ &\sigma = 2W + \delta(1 + c_* + c_*\eta), \qquad c_* = \frac{2}{1.0357712571566784} - \frac{4}{3}, \quad 0 < \delta \le 0.002. \end{aligned} \tag{2}$$

The decimal is exact. Fix one background on the previously registered nonconstant regular branch [1,2]. In its exterior-cone chart,

$$\begin{aligned} &g = dy^2 + \rho^2\gamma, \quad \mathrm{Ric}\,\gamma = 3\gamma, \quad a = \rho'/\rho, \quad p = \phi', \\ &a' = -\rho^{-2} - p^2/3, \quad p' = U_{,\phi} - 4ap, \quad a_b = \sigma_b/6 > 0, \quad p_b = -\sigma_b'/2 \neq 0. \end{aligned} \tag{3}$$

We assume the branch property $B_b = p_b'/p_b + \sigma_b''/2 > 0$. This paper does not recertify the background branch or take a uniform zero-detuning limit.

Let $dr/r = dy/\rho$, with $r/y \to 1$ at the regular endpoint, and write

$$g = \Omega(s)^2 \eta_{AB} dX^A dX^B, \quad \phi = \Phi(s), \quad s = R^2 - T^2, \quad f = \Phi_s, \quad b = r_b, \quad \rho_b = b\Omega_b. \tag{4}$$

The **actual Einstein–scalar background** is assumed analytically continued on $-s_- < s \le b^2$, with $\Omega > 0$ and $f \neq 0$. The prior regular-cone calculation explains this local continuation and its operator identities [1]. An arbitrary extension of coefficient functions would not establish the background equations.

The result is a sufficient family of coupled linearized solutions. For a fixed finite set of S^3 harmonics, arbitrary smooth radial data supported in a sufficiently near-shell annulus evolve on an explicit slab. Selected pure displacement and pure velocity states make actual shell contact within that slab. For $l \ge 2$ the resulting nonzero states are not smooth pure gauge. On a sufficiently thin fixed annulus the actual physical pairing is weakly nondegenerate; selected displacement–velocity pairs both contact the shell and retain a nonzero total pairing. Scalar uniqueness is proved for this family. Completeness of gravitational perturbations and uniqueness in a completed gauge quotient are separate questions.

2. Exact regular master and shell laws

Define $E = X^A \partial_A$ and $L = \Omega_s/\Omega + 2f_s/f$. The regular master and reconstruction from [1] are

$$\mathcal{L}F = \Box_\eta F - 2LEF - [8L + (8/3)sf^2]F = 0, \qquad P = r^4 F,$$

$$h_{AB} = 2F(s\eta_{AB} - 3X_A X_B), \qquad \chi = -\frac{3}{2\Omega^2 f}(E+4)F. \tag{5}$$

For arbitrary smooth F, the complete bulk residuals factor as

$$\delta\mathscr{E}_{AB} = \Omega^{-2}(3X_A X_B - s\eta_{AB})\mathcal{L}F, \qquad \delta\mathscr{S} = -\frac{3}{2\Omega^4 f}(E+6)\mathcal{L}F. \tag{6}$$

These differential identities prove sufficiency without dividing by a tangential harmonic eigenvalue and remain smooth at the cone and vertex.

With $A = (\Omega f^2)^{-1}$ and $\psi = \sqrt{A}F$, the canonical equation is

$$(\Box_\eta - V_c)\psi = 0, \qquad V_c = 3L - 2sL_s + sL^2 + (8/3)sf^2. \tag{7}$$

The shell is $R_b(T) = \sqrt{b^2 + T^2}$. Its exact master laws are

$$n_\eta\psi - c\Box_{\Sigma,\eta}\psi + d_e\psi = 0, \qquad n_\eta\psi + d_o\psi = 0, \qquad c = (\Omega_b B_b)^{-1} > 0,$$

$$d_j = \Omega_b\gamma_j + bL_b, \quad \gamma_e = 4/\rho_b - 4/(\rho_b^2 B_b), \quad \gamma_o = 4/\rho_b + p_b^2/(3a_b). \tag{8}$$

The first law is even and the second odd under exchange of the interiors. The shifted coefficients d_j need not be positive. Positive c is an auxiliary scalar boundary inertia, not an independently added physical shell field or a conclusion about the gravitational kinetic form.

3. Fixed hemisphere and controlled physical domain

Introduce $0 \le \vartheta \le \pi/2$, $|\tau| < \pi/2$, and

$$Q = \frac{b}{\cos\tau + \cos\vartheta}, \qquad T = Q\sin\tau, \quad R = Q\sin\vartheta,$$

$$\eta = Q^2[-d\tau^2 + d\vartheta^2 + \sin^2\vartheta\, d\Omega_3^2], \quad s = b^2\frac{\cos\tau - \cos\vartheta}{\cos\tau + \cos\vartheta}. \tag{9}$$

The shell is the equator of $M = S^4_+$, with $Q_b = b\sec\tau$ and boundary $\Gamma = S^3$. Choose

$$0 < \tau_* < \min\{\pi/4,\, 2\arctan(\sqrt{s_-}/b)\}, \qquad 0 < a_0 < d_0 < \tau_*/2. \tag{10}$$

The whole rectangle lies in the actual background because $-b^2\tan^2(\tau_*/2) \le s \le b^2$. It contains every point of the full ambient moving-ball slab

$$|T| \le T_0 = b\tan(\tau_*/2), \qquad R \le R_b(T). \tag{11}$$

Indeed $\tau = \arctan((T+R)/b) + \arctan((T-R)/b)$ is maximal at $R = 0$ for fixed positive T. The shell itself reaches $T = b \tan \tau_*$. These are different coverage bounds; a hemisphere rectangle is not an arbitrary global ambient-time slab.

Set $\psi = Q^{-3/2} u$. Direct differentiation gives

$$u_{\tau\tau} - \Delta_M u + \mathcal{V} u = 0, \qquad \mathcal{V} = 9/4 + Q^2 V_c(s). \tag{12}$$

The curvature term is the five-dimensional conformal coupling 3/16 times the cylinder scalar curvature 12. Equation (8) becomes

$$\partial_\nu u + \partial_\tau(\mu u_\tau) - \mu \Delta_\Gamma u + \beta u = 0, \quad \mu = \frac{c \cos\tau}{b} > 0,$$
$$\beta = Q_b(d_e - 3/(2b)) - \mu[(3/2)\sec^2\tau + (3/4)\tan^2\tau], \tag{13}$$

or $\partial_\nu u + \alpha_o u = 0$, with $\alpha_o = Q_b(d_o - 3/(2b))$. Here $\nu = \partial_\vartheta$. All undifferentiated coefficients are time even; μ' is odd.

For reference, positive control of smooth even solutions follows from

$$\mathcal{E} = \tfrac{1}{2}\int_M (u_\tau^2 + |\nabla u|^2 + m^2 u^2) + \tfrac{1}{2}\int_\Gamma [\mu(z_\tau^2 + |\nabla z|^2) + k z^2], \qquad z = u|_\Gamma,$$
$$\mathcal{E}' = \int_M (m^2 - \mathcal{V}) u u_\tau + \tfrac{1}{2}\int_\Gamma \mu'(|\nabla z|^2 - z_\tau^2) + \int_\Gamma (k - \beta) z z_\tau. \tag{14}$$

For $m, k > 0$, bounded coefficients and $\mu_{\min} > 0$ give $|\mathcal{E}'| \le C\mathcal{E}$. In odd parity retain the actual boundary potential $\frac{1}{2}\int_\Gamma \alpha_o z^2$ and choose a sufficiently large bulk mass. The epsilon trace inequality then gives coercivity and controls $\alpha_o' z^2$. An independently shifted boundary potential cannot be estimated by assuming an L^2 trace of the bulk velocity. Neither energy is asserted to be the physical Einstein–scalar Hamiltonian.

4. Smooth construction and actual contact

Fix $-\Delta_{S^3} H_l = \lambda_l H_l$, $\lambda_l = l(l+2)$, and put $x = \pi/2 - \vartheta$. Take smooth initial profiles supported in $[a_0, d_0]$. Such data vanish on an open shell neighborhood and therefore satisfy every corner compatibility condition: recursively differentiate (12) to obtain the bulk time jets and insert them into each time derivative of (13) or the Robin law. All these jets vanish at the boundary. The full hierarchy is recorded in the companion existence proof.

Writing $u = u_l(\tau, x) H_l$ and $v = \cos^{3/2} x \, u_l$ gives

$$v_{\tau\tau} - v_{xx} + W_l v = 0, \qquad W_l = Q^2 V_c + (\lambda_l + 3/4)\sec^2 x. \tag{15}$$

Let $p = v_\tau + v_x$, $m = v_\tau - v_x$, and $z = v(\tau, 0)$. For initial data (v_0, v_1), set $p_0 = v_1 + \partial_x v_0$ and $m_0 = v_1 - \partial_x v_0$. Collar support gives $z(0) = z'(0) = 0$. Then $p_\tau - p_x = -W_l v$ and $m_\tau + m_x = -W_l v$. The exact incoming boundary prescription is

$$\mu z'' + (\mu' + 1) z' + (\mu \lambda_l + \beta) z = p(\tau, 0), \qquad m(\tau, 0) = 2z' - p(\tau, 0), \tag{16}$$

or, in odd parity, $z' = p - \alpha_o z$ and $m = p - 2\alpha_o z$. Thus the dynamic boundary is a positive-mass ODE driven by the outgoing characteristic, not two freely prescribed incoming data.

For $t = \tau \ge 0$, the construction starts from

$$p(t, x) = p_0(x + t) - \int_0^t W_l(s, x + t - s) v(s, x + t - s)\, ds. \tag{17}$$

Integrate the other characteristic from the initial slice when $x \ge t$ and from (16) when $x < t$, then recover $v(t, x) = v_0(x) + \frac{1}{2}\int_0^t (p + m)(s, x)\, ds$. Together with the integrated boundary ODE these are Volterra equations. After substituting (17), their map on (v, z, z') has Lipschitz constant $C_l \Delta t$ in the uniform norm on a short interval; C_l depends only on bounded coefficients, $\mu_{\min}^{-1}$ and fixed λ_l. The odd case uses (v, z). Contraction gives uniqueness and existence. Successive differentiation gives smoothness; the zero initial corner neighborhood

removes all matching jumps along its outgoing characteristic. The characteristic constraints $v_x = (p-m)/2$ and $v(t,0) = z(t)$ propagate from the initial data.

The interval length depends on coefficient bounds, not solution amplitude, so iteration covers $|\tau| \le \tau_*$. The characteristic equations imply

$$\operatorname{supp} v(\tau,\cdot) \subset [\max(0, a_0 - |\tau|),\ d_0 + |\tau|]. \tag{18}$$

Since $d_0 + \tau_* < \pi/2$, no signal reaches the polar origin. Smoothly extending the coefficient outside this causal collar is only a construction device; the solution is zero before that extension can affect it. Extension by zero near the origin yields a smooth full-hemisphere solution. Finite linear combinations of harmonics follow by superposition. No uniform $l \to \infty$ estimate or general smoothness theorem from Galerkin compactness is presumed.

Finite propagation alone proves only that the shell trace is zero for $|\tau| < a_0$. Actual activation needs another argument. For nonzero displacement with support infimum exactly a_0 and zero velocity, time reversal makes v even. If its trace were zero on $(0,\tau_1)$, $\tau_1 > a_0$, it would vanish on $(-\tau_1,\tau_1)$ and the boundary law would give $v_x = 0$ there. Regarding (15) as the ordinary one-dimensional wave equation $v_{xx} - v_{\tau\tau} = W_l v$ with x as time, zero Cauchy data at $x = 0$ imply $v = 0$ in $x + |\tau| < \tau_1$. This contradicts an initial profile with support infimum a_0. For zero displacement and nonzero velocity, the solution is time odd and the same argument forces $v_\tau(0,x) = 0$, giving the same contradiction. This sideways argument is specifically one dimensional; it is not a general higher-dimensional timelike Cauchy theorem.

Thus either choice has nonzero shell trace somewhere in every interval immediately after a_0. No nonzero value or jet at exactly the smooth support edge is claimed. In ambient time,

$$T_c = b\tan a_0 = \frac{b^2 - R_0^2}{2R_0} < T_0, \qquad R_0 = b\tan(\pi/4 - a_0/2). \tag{19}$$

For prescribed master data, set $w = (\Omega f^2)^{1/2}$ and $Q_0 = b/(1+\cos\vartheta)$. On $T = \tau = 0$ the exact conversion is

$$u_0 = Q_0^{3/2} w^{-1} F_0, \qquad u_1 = Q_0^{5/2} w^{-1} (F_T)_0. \tag{20}$$

The weights are smooth and positive. Therefore both master states $(q_0 H_l, 0)$ and $(0, q_0 H_l)$ have the same contact infimum, where a_0 is the infimum of the nonzero profile's support in x.

Optionally imposing $d_0 + 2\tau_* < \pi/2$ keeps the whole causal support in $\vartheta > |\tau|$, hence $s > 0$. Contact can therefore occur entirely within the original regular positive-s collar. These data are a new family, not a continuation of the earlier near-cone nonzero-flux pair.

5. Full junction sufficiency

Scalar existence becomes a coupled result only after checking the complete shell system. In the exterior chart put $\Psi = P/\rho^2$; reconstruction implies, for arbitrary smooth packets,

$$\Psi' + 2a\Psi = -p\chi/3, \qquad p\chi' = p'\chi + 3\rho^{-2}(\Box_\gamma + 4)\Psi - 2p^2\Psi. \tag{21}$$

For normal graphs $y_i = y_b + \zeta_i$ in a common shell chart, direct linearized graph geometry gives

$$\delta h^{\mathrm{ind}}_{\mu\nu} = 2\rho_b^2(\Psi + a\zeta)\gamma_{\mu\nu}, \quad \delta\phi^{\mathrm{ind}} = \chi + p\zeta,$$

$$\delta K^\mu{}_\nu = (\Psi' + 2a\Psi + a'\zeta)\delta^\mu{}_\nu - \rho^{-2} D^\mu D_\nu \zeta, \quad \delta p_n = \chi' + 2p\Psi + p'\zeta. \tag{22}$$

The graph evaluation and lapse terms are essential. Write $P_\pm = P_e \pm P_o$ and choose

$$\zeta_e = 0, \qquad \zeta_o = -P_{ob}/(a_b\rho_b^2), \qquad \zeta_\pm = \pm\zeta_o. \tag{23}$$

Induced metric continuity is immediate; scalar continuity is precisely $P_o' = -p_b^2 P_o/(3a_b)$, the odd law. The full trace-reversed Israel residual and summed scalar residual reduce to

$$\delta\mathcal{R}^\mu{}_\nu = -2\rho_b^{-2}(D^\mu D_\nu + \delta^\mu{}_\nu)\zeta_e,$$

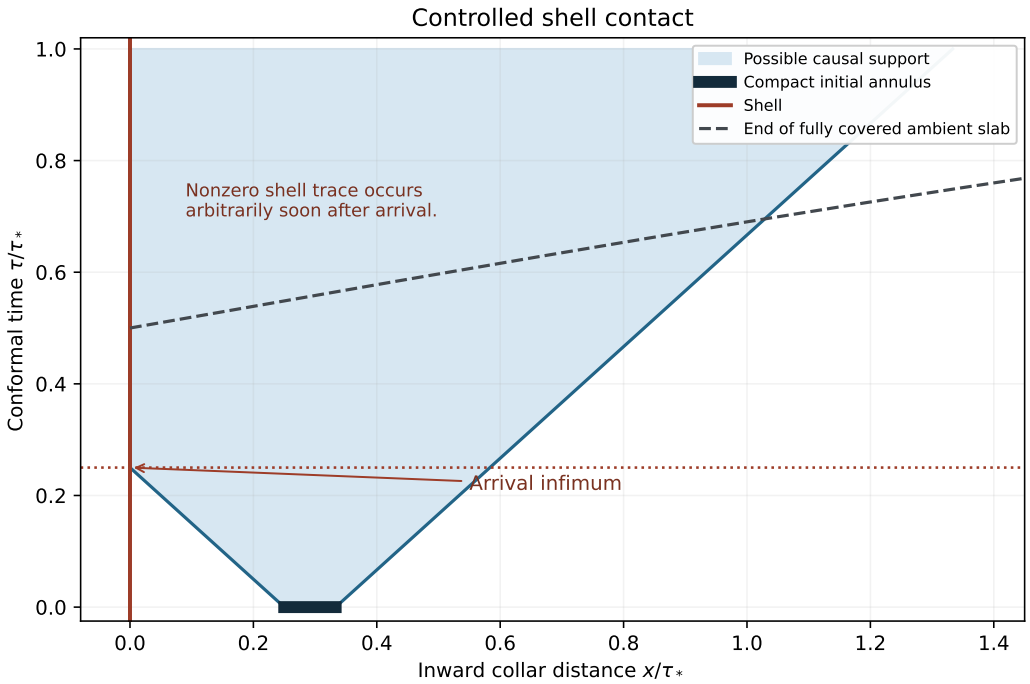


Figure 1: Possible causal support for illustrative collar data, not an evolved waveform. The trace activates arbitrarily soon after the arrival infimum; smooth data remain flat at the exact front. The dashed curve bounds the fully covered ambient-time slab. Its scale is illustrative and does not certify a background continuation.

$$\delta\mathcal{S}_\Sigma = -\frac{6}{\rho_b^2 p_b}[B_b P_e' - \rho_b^{-2}(\Box_\gamma + 4)P_e] + 2p_b B_b \zeta_e. \tag{24}$$

Both vanish by (23) and the even law. This checks all tensor components, not merely the trace-free part. The difference of the bulk Hamiltonian constraints supplies normal-force consistency: $3a_b\delta K_o = p_b \delta p_{n,o}$, equivalently $\sigma_b \delta K_o + \sigma_b' \delta p_{n,o} = 0$. Codazzi gives tangential conservation. There is no extra odd wave equation to impose.

These are sufficiency statements without harmonic division. Necessity for all embeddings is different: Israel permits $(D_\mu D_\nu + \gamma_{\mu\nu})\zeta_e = 0$, with the five de Sitter ambient-coordinate solutions. The scalar condition then still contains ζ_e. Setting it to zero selects a sufficient family and does not classify these exceptional representatives.

6. Physical current and nonzero conserved pairing

Fix the action-derived potential convention

$$\theta^A = \sqrt{-g}\{\tfrac{1}{2}(\nabla_B h^{AB} - \nabla^A h) - \nabla^A \phi\, \chi\}, \qquad \omega^A = \delta_1\theta^A(2) - \delta_2\theta^A(1).$$

For one interior the exact reconstructed currents are

$$\omega^y = 9\sqrt{-\gamma}\,\mathfrak{a}[((\Box_\gamma + 4)P_1)P_2' - ((\Box_\gamma + 4)P_2)P_1'], \qquad \mathfrak{a} = (\rho^2 p^2)^{-1},$$
$$\omega^\mu = \sqrt{-\gamma}\rho^2[\chi_1 D^\mu \chi_2 - \chi_2 D^\mu \chi_1 + 6(\Psi_1 D^\mu \Psi_2 - \Psi_2 D^\mu \Psi_1)]. \tag{25}$$

For two even states the boundary law makes $\omega_b^y = 0$ pointwise; there is no additional straight-even-shell current in this slice. For two odd states the completed graph and shell derivative action is

$$S^{(2)}_{\Sigma,\mathrm{der}} = -3a_b\rho_b^2 \int \sqrt{-\gamma}(D\zeta_o)^2 = -9c_o \int \sqrt{-\gamma}(DP_{ob})^2, \qquad c_o = (3a_b\rho_b^2)^{-1}. \tag{26}$$

It includes both GHY contributions, the Einstein tangential-divergence corner, the shifted bulk integration domains, and one common shell tension. Omitting the moving-domain term reverses the resulting coefficient. The companion junction calculation derives these terms independently.

Polarization gives the physical shell current and balance

$$\omega^\mu_\Sigma = 18c_o\sqrt{-\gamma}(P_{1b}D^\mu P_{2b} - P_{2b}D^\mu P_{1b}), \qquad \partial_\mu\omega^\mu_\Sigma = \omega^y_+(y_b) + \omega^y_-(y_b). \tag{27}$$

Mixed reflection parities cancel. In ambient polar coordinates the normal moving-boundary flux is $\omega^R - R_b'\omega^T = \omega^y/R_b$. Thus, for a full interior with no additional boundary, the future-oriented total pairing

$$\sigma_T = -\left[\sum_i \int_{R\le R_b(T)} \omega_i^T\, dR\, d\Omega + \int_{S^3} \omega^T_\Sigma\, d\Omega\right] \quad \text{satisfies} \quad \frac{d\sigma_T}{dT} = 0. \tag{28}$$

The overall minus sign specifies the future-Cauchy convention for the displayed vector densities; reversing the common orientation changes no conservation statement. The cone is internal to the smooth domain. A region artificially cut there must retain its cone flux separately.

At $T = 0$, let $(q, v) = (F, F_T)$ have compact support in an annulus inside the shell. The shell cut term is initially zero. Per interior, direct pullback of (25) yields

$$\sigma((q_1, v_1), (q_2, v_2)) = K(q_1, v_2) - K(q_2, v_1),$$

$$K(q, v) = \int dr\, d\Omega_3 \left[m_r(D_r + 4)q(D_r + 5)v + \frac{6r^7}{\Omega}qv\right], \quad D_r = r\partial_r, \quad m_r = \frac{9r^3}{4\Omega f^2} > 0. \tag{29}$$

The unit difference between the two first-order factors follows from $[\partial_T, E] = \partial_T$. Compact radial integration by parts gives

$$K(q, q) = \int dr\, d\Omega_3 \left[m_r\{(D_r q)^2 + (2 + 9sL)q^2\} + \frac{6r^7}{\Omega}q^2\right]. \tag{30}$$

For $I = (r_1, r_2)$, write $w_I = r_2 - r_1$, $m_* = \min_{\bar I}(m_r r^2) > 0$ and $M_* = \max_{\bar I} m_r(-2 - 9sL)_+$. Dirichlet Poincaré gives

$$K(q, q) \ge \left(\frac{\pi^2 m_*}{w_I^2} - M_*\right) \int_I dr\, d\Omega_3\, q^2. \tag{31}$$

Every regular near-shell collar contains an annulus thin enough for the bracket to be positive while satisfying (10). Hence two same-reflection- parity states $(q, 0)$ and $(0, q)$ have doubled physical pairing $2K(q, q) > 0$. Section 4 proves actual contact for both; (28) conserves their nonzero **total** pairing afterward, including the odd cut term. Nonzero boundary traces do not establish a nonzero local shell-current Wronskian, and no such claim is made.

On all smooth data supported in this fixed thin annulus with a fixed finite harmonic set, the pairing is weakly nondegenerate. A kernel element (q, v) tested against $(0, q)$ has $q = 0$ by (31); testing next against $(v, 0)$ gives $v = 0$. Smooth uniqueness and (28) transport this statement to the solution image. K need not be symmetric. Positivity of its diagonal is not a physical energy theorem or a completed-space identification.

7. Gauge obstruction and scope

For $l \ge 2$, non-gauge character also follows directly on a polar annulus. Write $g = g_{ab}(z)dz^a dz^b + w(z)^2 q_{ij} d\omega^i d\omega^j$ and the scalar part of a gauge covector as $\xi_a = f_a H_l$, $\xi_i = J D_i H_l$. Then

$$(\mathcal{L}_\xi g)^{\mathrm{TF}}_{ij} = 2J(D_i D_j H_l)^{\mathrm{TF}}, \quad (\mathcal{L}_\xi g)_{ai} = [\partial_a J - 2(w_a/w)J + f_a]D_i H_l,$$

$$\int |(\mathrm{Hess}\, H_l)^{\mathrm{TF}}|^2 = \tfrac{2}{3}\lambda_l(\lambda_l - 3)\int |H_l|^2 > 0. \tag{32}$$

The reconstruction has angular trace only and zero mixed component, so $J = 0$ and then $f_a = 0$. Orthogonal angular sectors cannot cancel this scalar projection. A nonzero reconstructed $l \ge 2$ state is therefore not smooth pure gauge. This does not classify $l = 0, 1$, finite-cut charges or weaker gauge groups.

The result combines four separate ingredients: actual-background bulk reconstruction, smooth finite-harmonic boundary evolution, complete junction sufficiency, and the physical current including shell motion. A scalar boundary theorem alone would not imply the other three. General angular Sobolev completion, unrestricted initial-boundary wellposedness, all-sector completeness, nonlinear integrability and stability remain open here. The construction gives no reheating mechanism, observational discrimination or evidence for a higher-dimensional origin of the Big Bang.

8. Prior art, verification and declarations

Positive dynamic-boundary inertia, scalar compatibility and energy methods are established subjects. Zahn [3] treats static flat generalized Wentzell boundaries; Vitillaro [4] proves nonlinear energy and stronger-regularity results on fixed bounded domains. Marta [5] studies time-dependent compact Riemannian metrics. These are precedents, not already verified theorems for the present moving shell and coupled reconstruction. We use our own finite- coefficient Gronwall bound and explicit characteristic proof.

The recent Robin treatment of Costeri, Dappiaggi and Juárez-Aubry [6] provides smooth interior-compact Cauchy existence and causal support on spacetimes with timelike boundary. Its inward-normal convention maps to $\kappa = -d_o$. Extending its stated curvature-mass family to V_c requires an explicit smooth lower-order adaptation. Its later microlocal reflection hypotheses are not silently added to this finite-slab smooth construction. The accompanying literature register records primary-source reading and failed searches. It supports a bounded comparison, not external or worldwide originality. No new general scalar boundary theory is claimed.

The package contains detailed existence, junction/current and independent audit notes, plus original symbolic programs and strict wrong-formula controls. They check coordinate and conformal transformations, full graph geometry, arbitrary-packet residuals, moving-shell coefficients, and the annular Cauchy identity. The explicit contracts contain **84 positive algebraic identities and 14 wrong-formula controls per execution mode**: 21/4 for the coordinate checker and 63/10 for junctions and currents. Normal and optimized individual outputs agree byte for byte. The portable replay additionally checks relocation, input integrity, receipt equality and injected failures; its receipts record the actual execution status. Algebraic check counts are not a count of independent physical results or a machine-certified PDE theorem.

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AI assistance and review. Substantial OpenAI Codex assistance was used for derivation, symbolic programming, literature organization, internal checking and drafting. Separate agent processes supplied internal reviews; these are not external human peer review. The author retains responsibility for verification, interpretation and any submission. APS submission and external novelty assessment remain open; neither acceptance nor readiness for acceptance is asserted.

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Companion proof files. Constructive existence; junctions and physical current; independent energy audit; bounded primary-source comparison.